04 / Grounding

What connects the representation or interpretation to physical evidence?

EX06 / NeuralFeels | Source analysis | C1 | P: paper; W: project; R: repository

Prediction / representation	Matched reference	Difference / ambiguity	Conclusion / scope limit

1. Comparison boundary

Name the prediction/measurement and reference. Match conditions; identify simulated versus physical evidence.

Compare vision-only and fusion on identical unknown-shape sequences. Mesh/reference tracker ground the estimates; known-CAD results form a separate experiment. [P 2.1-2.3]

2. Quantities and validity

Declare quantity, units, frame, tolerance and measurement limits. Distinguish a chosen tolerance from observed uncertainty.

ADD-S handles object symmetry; F-score uses a chosen 5 mm distance threshold. Physical multi-camera reference has uncertainty and shares visual information. [P 2.1; S1]

3. Results and uncertainty

Give paired outcomes, trial counts and uncertainty or limits. If untested, record the protocol and missing evidence explicitly.

Reported pose-drift improvement: 21.3% simulated, 26.6% physical versus vision-only. Fig 3 covers 40/30 sequences, five seeds each; do not pool origins. [P 2.2]

4. Interpretation and challenge

Bound the conclusion and consider another cause of the difference. Choose a diagnostic test or scope restriction.

Reference dependence can bias small differences. Compare both modalities against the same reference and inspect occluded/failed sequences; relative improvement alone is not absolute accuracy. [P Fig 6; S1]

Review: Could sensing error, reset differences or selection explain the discrepancy instead of model error?

Example response: The physical reference shares visual information; compare modalities against the same reference and disclose uncertainty.

Worksheet 4 / Grounding - evidence and team review

EX06 / NeuralFeels | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX06 Project: NeuralFeels Claim ID: C1 Revision: draft 2

Mode: Source analysis Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: selected paper text/tables checked 20 Oct 2026; no raw dataset download or independent experiment.

Evidence ID	Artifact/source + location (clickable)	What it supports; limitation / access gap
P	arxiv.org/html/2312.13469v1	Paper: method/results; reported evidence, not our measurement.
W	suddhu.github.io/neural-feels	Project: system/videos; selected examples do not establish reliability.
R	github.com/facebookresearch/neuralfeels	Repository: linked route; code, raw data and current commit not audited.

Next action / owner / verification:

Analyst: compare modalities on identical sequences/reference; record 21.3/26.6 percent improvements and reference uncertainty.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

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P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.